

Weathering

As soon as rock is exposed above sea level, it comes under attack from the weather. Rain is slightly acidic and eats away at all kinds of rocks, from soft limestone to the toughest granite. Acids in the soil or made by plants also contribute to this process. Meanwhile, frost and baking sun gradually loosen fragments that are then carried away by the processes of erosion.

BREAKDOWN

Water vapor dissolves carbon dioxide from the air to form weak carbonic acid that falls as rain. The acid breaks down certain minerals. For example, the feldspar minerals in granite are turned into soft clay, releasing tougher quartz crystals as sand grains. Over time, the rock is weathered on the outside, becoming rounded as the edges crumble away.



ACID ATTACK

Limestone is slowly dissolved by acid rainwater. The water eats away the surface, enlarging cracks to form limestone pavements, such as these in Yorkshire, England. As the water seeps through fissures, it also creates limestone caves. In hot climates, the process is sped up, destroying most of the limestone and leaving a few rocky pinnacles.



MICROBES AND ROOTS

Living things also attack rocks. Microbes break down rocks so they can use their minerals as nutrients—all living things need elements such as iron and potassium, which can be found in rocks. In the same way, the lichens that grow on rocks use acids to dissolve them. Plant roots penetrate rocks and pry them apart as the roots grow—especially tree roots, which become broader with age.



► PROBING ROOTS

These tree roots are not just clinging to the rock—they are also dissolving its minerals.